AeroPak Airlines — Sales & Ticketing Analysis

Power BI analysis · 53,530 bookings · January 2023 – December 2024

1. Executive Summary

AeroPak sold 53,530 tickets over 2023–2024, about 73 a day, and 91.5% of them were confirmed. Domestic flights account for 57.5% of bookings, but international routes earn the revenue: every top-10 revenue route is international, led by Lahore–New York ($1.40M). First class brings in the most fare revenue ($15.5M) and Business the least ($10.1M), even though both have the same 20% share of bookings. Demand is steady across both years, and bookings are spread almost evenly across all five channels and payment methods. The main risk to reporting is data quality. Cancelled and refunded tickets record fares inconsistently, which overstates revenue unless they are filtered out.

2. Data & Approach

- **Source:** a raw ticketing-system export with three tables: ticket sales (~54,000 rows), customers (~9,000) and airports (26).
- **Cleaning:** Power Query, covering text standardization, currency parsing, deduplication and checks for customer IDs with no match.
- **Model:** a star schema with Fact_TicketSales at the center and Dim_Customer, Dim_Airport and Dim_Date around it. Dim_Date was built in Power Query and marked as the date table.
- **Reporting:** a 3-page Power BI dashboard (Revenue & Routes, Customer & Channel, Status & Trends).

3. Data Cleaning & Assumptions

- **Text standardization:** BookingChannel, TravelClass and FlightType had mixed casing (e.g. “CALL CENTER”, “call center”). These were trimmed and set to Capitalize Each Word, and “Intl” was replaced with “International”.
- **Currency:** FareAmount mixed plain numbers with “$”-prefixed and “USD”-prefixed text. Symbols and commas were removed and the column converted to decimal. All fares are assumed to be in USD.
- **Rounding:** floating-point noise (e.g. 78.43000000000001) was rounded to 2 decimals.
- **Airport codes:** lowercase codes (e.g. “lhe”, “yyz”) were converted to uppercase to match Dim_Airport.
- **Duplicates:** a duplicate is the same TicketID appearing more than once. Duplicate rows were removed, keeping the first occurrence.

  Unmatched customers: rows using CustomerID CUS-99999, which doesn’t exist in Dim_Customer and appears to be a placeholder, were removed.
- **Ancillary revenue:** zero values were kept as genuine (the customer bought no extras).

- **Revenue definition:** fare revenue is FareAmount only, excluding tax and ancillary. FareAmount is treated as the final amount charged.
- **Confirmed bookings only:** revenue and average-fare visuals use Confirmed bookings only. Cancelled and refunded fares aren't earned revenue, and the system records them inconsistently. Some are 0 and others keep the full fare; for example, TKT0011879 is Cancelled but shows $643.45.
- **Dates:** monthly trends use BookingDate.
- **Relationships:**
    - All relationships are one-to-many with single-direction filtering.
    - Origin airport and BookingDate are active. Destination airport and FlightDate are inactive.
    - A Route column (Origin → Destination) was added in Power Query so routes can be analyzed without the inactive relationship.

4. Business Questions Answered

Revenue & Route Performance (Page 1)

Question	Answer	Visual
Which routes have the most bookings?	LHE → SIN (1,018), ISB → KDU (1,002), then KHI → KDU and LHE → YYZ (994 each). The top 10 range from 962 to 1,018, so no route dominates.	Top 10 Routes by Number of Bookings
Which routes earn the most fare revenue?	LHE → JFK ($1.40M), KHI → MAN ($1.36M), then KHI → KUL and LHE → YYZ ($1.35M each). All top 10 are international.	Top 10 Routes by Total Fare Revenue
How does fare revenue compare across classes?	First $15.5M, Economy $11.1M, Business $10.1M (about $36.7M in total).	Total Fare Revenue by Travel Class
How do domestic and international bookings compare?	Domestic 30.79K (57.5%), International 22.75K (42.5%).	Bookings by Flight Type
What is the average fare by class?	First $1,598, Business $1,029, Economy $377.	Average Fare by Travel Class

Customer & Channel Behavior (Page 2)

Question	Answer	Visual
Which booking channel is used most?	Airport Counter and Mobile App, tied at 10,745 bookings each. All five channels fall between 10,644 and 10,745, within 1% of each other.	Bookings by Booking Channel
How does class selection vary by channel?	It doesn't. Every channel is about 60% Economy, 20% Business and 20% First.	Travel Class Selection by Booking Channel
What is total ancillary revenue?	$1.28M.	Total Ancillary Revenue card
Which payment method is used most?	Wallet, with 10,742 bookings (20.07%), just ahead of Debit Card and Bank Transfer (20.05% each). All five methods fall between 19.9% and 20.1%.	Bookings by Payment Method
How many unique customers booked at least once?	About 8,980, nearly all of the ~9,000 customers on file	Unique Customers card

Booking Status & Trends (Page 3)

Question	Answer	Visual
How many bookings fall into each status?	Confirmed 48,958 (91.5%), Cancelled 2,507 (4.7%), Refunded 2,065 (3.9%).	Bookings by Status
What share was cancelled or refunded?	8.54% (4,572 of 53,530 bookings).	Cancelled & Refunded card; Bookings by Status
How does booking volume trend by month?	Flat, between about 2,050 and 2,320 a month (about 73 a day). The lower months are the shorter ones, February and the 30-day months, not months of weaker demand.	Monthly Booking Trend
Which month had the most bookings?	**January 2024** with **2,314** bookings.	Monthly Booking Trend
How does revenue trend month to month?	It varies between about $1.43M and $1.66M a month, with no clear growth over the two years.	Monthly Revenue Trend

5. Key Findings

1. **International routes carry the revenue.** They are 42.5% of bookings, yet all ten top revenue routes are international, each earning between $1.29M and $1.40M.
2. **Business class underperforms.** It has the same ~20% booking share as First but earns $5.4M less. It also trails Economy, which wins on volume (60% of bookings) despite an average fare about a third of Business.
3. **Demand is stable.** Bookings hold at about 73 a day across both years. Month-to-month swings mostly follow the length of the month rather than seasonal demand.
4. **No channel or payment method stands out.** Each takes about 20% of bookings, and the class mix is the same in every channel, so booking volume alone can't guide channel strategy.
5. **Cancelled and refunded fares are recorded inconsistently.** 8.54% of bookings are cancelled or refunded, and some still carry a full fare. Any report that doesn't filter by status will overstate revenue.

6. Recommendations

1. **Fix how cancellations and refunds are recorded.** Zero out or flag the fare on every cancelled or refunded ticket, and store refund amounts in a separate field. Revenue reports will then be accurate without manual filtering.
2. **Grow Business class on international routes.** Business has the same booking share as First but earns $5.4M less. Test targeted upgrade offers on the top revenue routes (LHE → JFK, KHI → MAN, LHE → YYZ). Each Economy-to-Business upgrade raises the average ticket from $377 to $1,029.
3. **Enforce clean data at the point of sale.**
    - Block bookings without a valid CustomerID; CUS-99999 appears to be a default placeholder.
    - Use fixed drop-down values for channel, class and flight type so the inconsistencies cleaned here don't come back.

7. Next Steps

- Break down the 8.54% cancellation and refund rate by channel, class and route to find where it is concentrated.
- Analyze revenue by destination city and country by switching on the inactive Destination relationship with DAX.
- Add booking lead time (days between booking and flight) to see how far ahead customers buy and whether it affects fares.
